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Chapter-1 INTRODUCTION

With the development technologies in the areas of augmented reality and devices that we use in our daily life, these devices are becoming compact in the form of Bluetooth or wireless technologies. This paper proposes an AI virtual mouse system that makes use of the hand gestures and hand tip detection for performing mouse functions in the computer using computer vision.

The main objective of the proposed system is to perform computer mouse cursor functions and scroll function using a web camera or a built-in camera in the computer instead of using a traditional mouse device.

Hand gesture and hand tip detection by using computer vision is used as a **HCI (Human Computer Interaction)** with the computer. With the use of the AI virtual mouse system, we can track the fingertip of the hand gesture by using a built-in camera or web camera and perform the mouse cursor operations and scrolling function and also move the cursor with it.

While using a wireless or a Bluetooth mouse, some devices such as the mouse, the dongle to connect to the PC, and also, a battery to power the mouse to operate are used, but in this paper, the user uses his/her built-in camera or a webcam and uses his/her hand gestures to control the computer mouse operations. In the proposed system, the web camera captures and then processes the frames that have been captured and then recognizes the various hand gestures and hand tip gestures and then performs the particular mouse function.

Python programming language is used for developing the AI virtual mouse system, and also, **OpenCV** which is the library for computer vision is used in the AI virtual mouse system. In the proposed AI virtual mouse system, the model makes use of the **MediaPipe** package for the tracking of the hands and for tracking of the tip of the hands, and also, **Pynput, Autopy,** and **PyAutoGUI** packages were used for moving around the window screen of the computer for performing functions such as left click, right click, and scrolling functions.

The results of the proposed model showed very high accuracy level, and the proposed model can work very well in real-world application with the use of a CPU without the use of a GPU.

Chapter-2 PROFILE OF PROBLEM ASSIGNED

2.1 PROBLEM DEFINITION :-

This AI virtual mouse can be made use of in places where using a physical mouse is not feasible such as moist or wet conditions, to aid people who can't handle a device and where there isn't ample space for using a mouse and to eliminate the need of mouse and cable altogether.

In view of the pandemic, there arises a need to avoid high-contact surfaces, out of which the mouse is one. In order to eliminate the need to avoid contact, the system provides an intuitive way to interact with the computer system with the use of hand-gestures and finger tips to emulate mouse-like functions such as scrolling, clicking, and pointing.

2.2 PROBLEM STATEMENT :-

This system is designed to work on projects reducing the potential risks and helping to ensure that the project is delivered on time. I serve, As a repository for all the documents ,drawings and communications relating to the project and are used by all project participants to access, read, print and edit material according to authorization setup by the project administrator.

The essence of collaboration software is to develop a process whereby documents are all electronic, thus enabling them to be located at a Secure central location that can be accessed by those to whom access rights have been given.

2.3 OBJECTIVE :-

The main objective of the proposed AI virtual mouse system is to develop an alternative to the regular and traditional mouse system to perform and control the mouse functions, and this can be achieved with the help of a web camera that captures the hand gestures and hand tip and then processes these frames using image processing to perform the particular mouse function such as left click, right click, scrolling function and much more.

Chapter-3 SYSTEM REQUIREMENT

3.1 HARDWARE AND SOFTWARE REQUIREMENTS :-

3.1.1 Hardware requirements :-

1	Core i3 Processors
2	4 GB of RAM
3	5 GB of Hard Disk
4	Webcam
5	Client Machine

3.1.2. Software requirements :-

1	Operating system
2	Visual Studio Code
3	Python Version 3.6.0 and greater
4	Anaconda IDE

3.2 LANGUAGES USED :-

3.2.1. Python Programming Language:-

Python is a high-level, general-purpose and a very popular programming language. Python programming language (latest Python 3.6.0) is being used in web development, Machine Learning applications, along with all cutting edge technology in Software Industry. Python Programming Language is very well suited for Beginners, also for experienced programmers with other programming languages like C++ and Java.

Figure:- 3.2.1

Below are some facts about Python Programming Language:-

1. Python is currently the most widely used multi-purpose, high-level programming language.
2. Python allows programming in Object-Oriented and Procedural paradigms.
3. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.
4. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc.
5. The biggest strength of Python is huge collection of standard library which can be used for the following:-
   - ✓ Machine Learning
   - ✓ GUI Applications (like Kivy, Tkinter, PyQt etc.)
   - ✓ Web frameworks like Django (used by YouTube, Instagram, Dropbox)
   - ✓ Image processing (like OpenCV, Pillow)
   - ✓ Web scraping (like Scrapy, BeautifulSoup, Selenium)
   - ✓ Test frameworks
   - ✓ Multimedia
   - ✓ Scientific computing
   - ✓ Text processing and many more.

3.2.2. Libraries Used For AI Virtual Mouse:-

3.2.2.1. MediaPipe:-

MediaPipe is a framework which is used for applying in a machine learning pipeline, and it is an open source framework of Google. The MediaPipe framework is useful for cross platform development since the framework is built using the time series data. The

MediaPipe framework is multimodal, where this framework can be applied to various audios and videos.

The MediaPipe framework is used by the developer for building and analyzing the systems through graphs, and it also been used for developing the systems for the application purpose. The steps involved in the system that uses MediaPipe are carried out in the pipeline configuration.

The pipeline created can run in various platforms allowing scalability in mobile and desktops. The MediaPipe framework is based on three fundamental parts; they are performance evaluation, framework for retrieving sensor data, and a collection of components which are called calculators, and they are reusable.

A pipeline is a graph which consists of components called calculators, where each calculator is connected by streams in which the packets of data flow through. Developers are able to replace or define custom calculators anywhere in the graph creating their own application.

The calculators and streams combined create a data-flow diagram; the below graph is created with MediaPipe where each node is a calculator and the nodes are connected by streams.

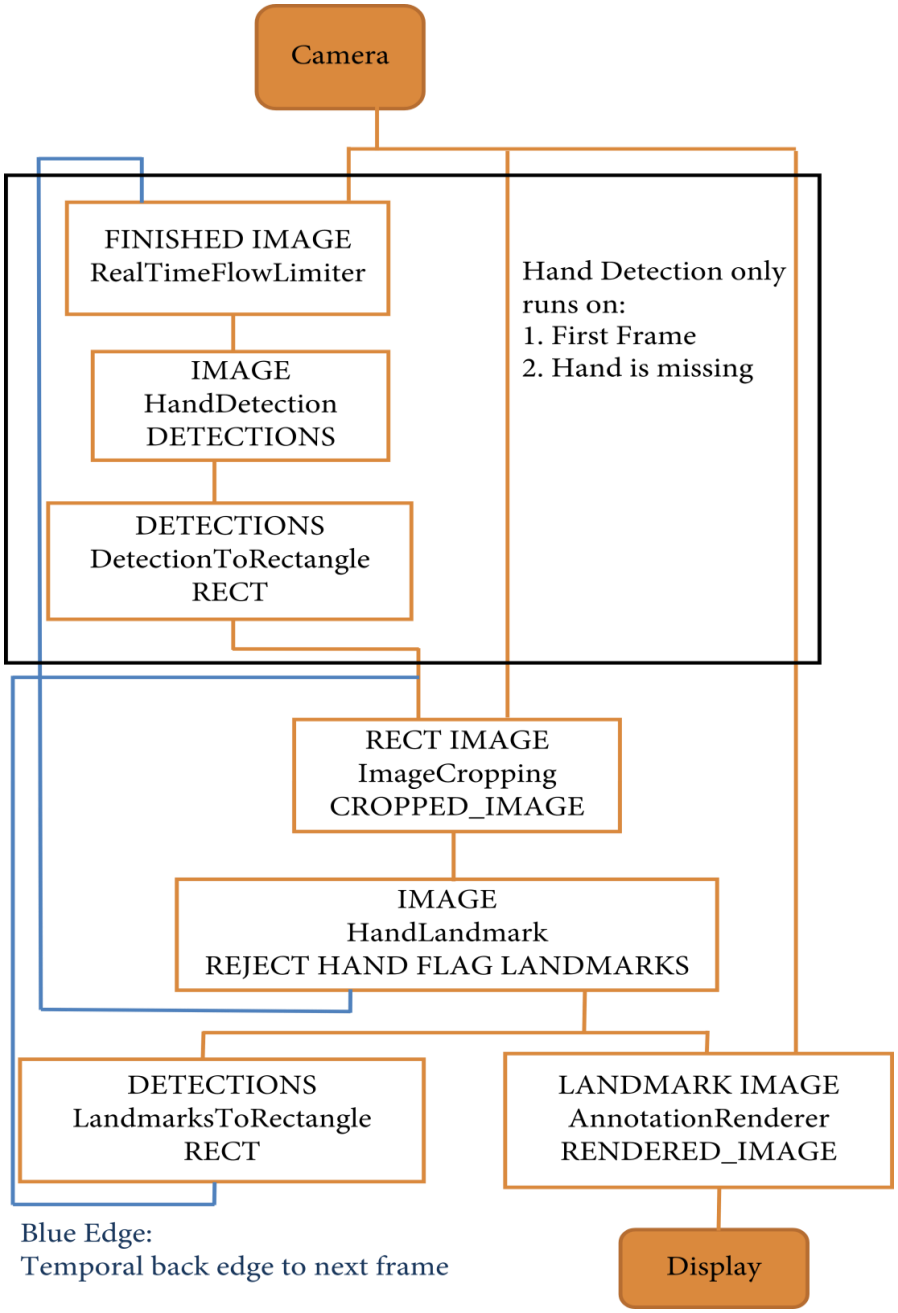
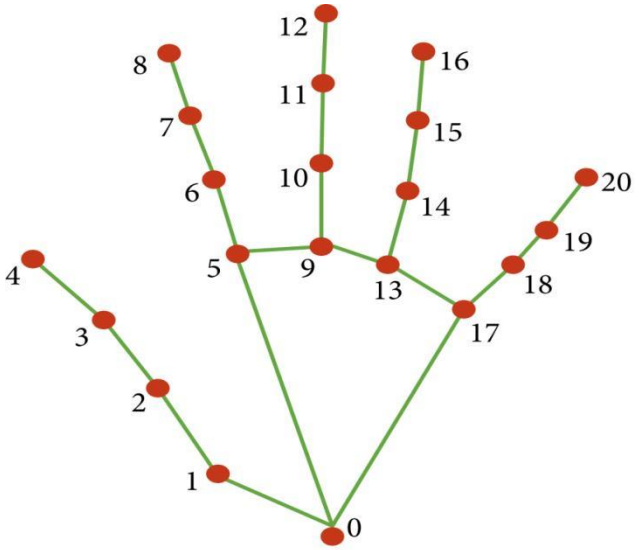


Figure :- 3.2.2.1

Single-shot detector model is used for detecting and recognizing a hand or palm in real time. The single-shot detector model is used by the MediaPipe. First, in the hand detection module, it is first trained for a palm detection model because it is easier to train palms. Furthermore, the nonmaximum suppression works significantly better on small objects such as palms or fists. A model of hand landmark consists of locating 21 joint or knuckle co-ordinates in the hand region, as shown in below diagram.



0. WRIST	11. MIDDLE_FINGER_DIP
1. THUMB_CMC	12. MIDDLE_FINGER_TIP
2. THUMB_MCP	13. RING_FINGER_MCP
3. THUMB_IP	14. RING_FINGER_PIP
4. THUMB_TIP	15. RING_FINGER_DIP
5. INDEX_FINGER_MCP	16. RING_FINGER_TIP
6. INDEX_FINGER_PIP	17. PINKY_MCP
7. INDEX_FINGER_DIP	18. PINKY_PIP
8. INDEX_FINGER_TIP	19. PINKY_DIP
9. MIDDLE_FINGER_MCP	20. PINKY_TIP
10. MIDDLE_FINGER_PIP	

Figure :- 3.2.2.1

3.2.2.2 OpenCV:-

OpenCV is a Python open-source library, which is used for computer vision in Artificial intelligence, Machine Learning, face recognition, etc.

Figure:- 3.2.2.2

In OpenCV, the CV is an abbreviation form of a computer vision, which is defined as a field of study that helps computers to understand the content of the digital images such as photographs and videos.

The purpose of computer vision is to understand the content of the images. It extracts the description from the pictures, which may be an object, a text description, and three-dimension model, and so on. For example, cars can be facilitated with computer vision, which will be able to identify and different objects around the road, such as traffic lights, pedestrians, traffic signs, and so on, and acts accordingly.

CHAPTER-4 FEASIBILITY ANALYSIS

Before the analysis is done, the feasibility study needs to be done so as to check whether the system is feasible to run on the machines or not. Each and every aspect was judged and after this study.

4.1 OPERATIONAL FEASIBILITY :-

Operational feasibility is a measure of how well a proposed system solves the problems, and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development. The operational feasibility assessment focuses on the degree to which the proposed development projects fits in with the existing business environment and objectives with regard to development schedule, delivery date, corporate culture, and existing business processes.

4.2 TECHNICAL FEASIBILITY :-

The various schedules were assigned keeping in mind the cost and performance of the system. The hardware requirements were taken into consideration so that the system runs on the proper platform. So it fulfills the technical feasibility. So, our project can run on any operating system like window 8,10 etc…so it is technically feasible.

4.3 ECONOMIC FEASIBILITY :-

Each and every system requires that the cost be taken into consideration. Our system too had considered this. Because, our project reduces the extra cost for the manual labour as well as costly software. That's why the system is considered best which is being produced with better efficiency with reasonable cost and we have tried to achieve this target so as to make our system efficient in every aspect.

4.4 BEHAVIORAL FEASIBILITY :-

My project can be changed according to the user's requirement at any time because it is dynamic in nature, so it fulfills the behavioral feasibility.

CHAPTER-5 PROJECT PLAN

Project planning establishes a plan to develop the project.It describes the technical tasks to be conducted, the risks that are likely, the resources that will be required, the work item to be produced , and a work schedule.

5.1 PROJECT SCHEDULE :-

The following steps were followed in developing the application:-

5.1.1 Study ,Analysis and Requirement Specifications :-

The study of the existing system and its drawbacks gave an idea about what kind of interface was required. The study was conducted through information gathering and studying the existing system.

5.1.2 Coding of the Desired System :-

A modular approach was followed, where , in each module a well-defined procedure was developed independently. For the coding of the project python programming language is used as stated above.

CHAPTER-6 DEVELOPMENT PHASES

Project Schedule is a list of activities that need to be performed in order to fulfill the project scope. This list is organized in a logical sequence, also called the schedule network logic. The list consists of activity duration, constraints, inter-dependencies among the various activities, required to complete the schedule activity. Project schedule specifies the planned start and finish dates of each activity likely to be executed in a project. The project schedule serves as a baseline against which the project progress can be tracked.

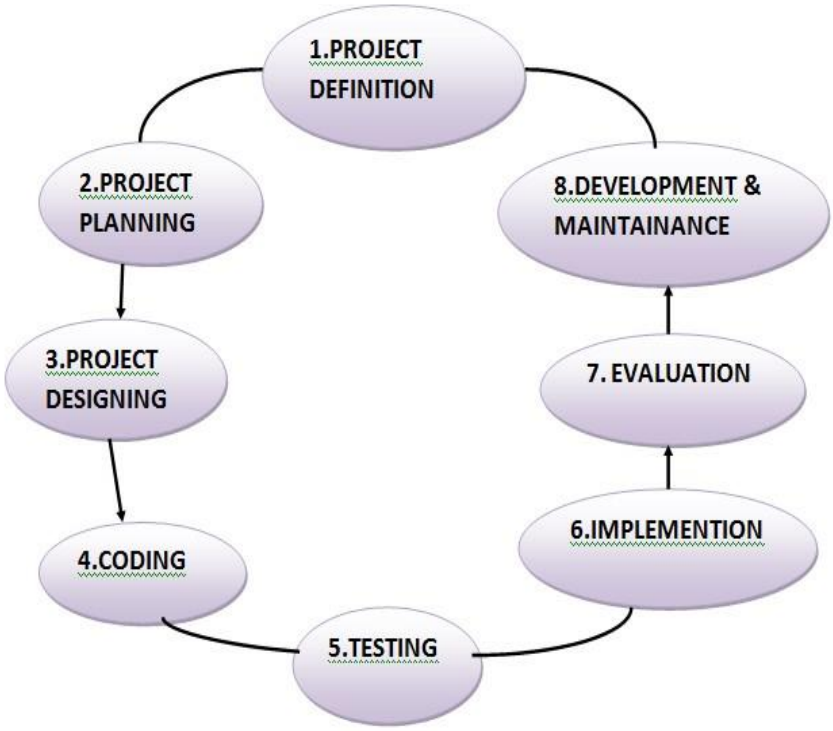


Figure:- Development Phases

CHAPTER-7 DATA FLOW DIAGRAM (DFD)

7.1 DATA FLOW DIAGRAM :-

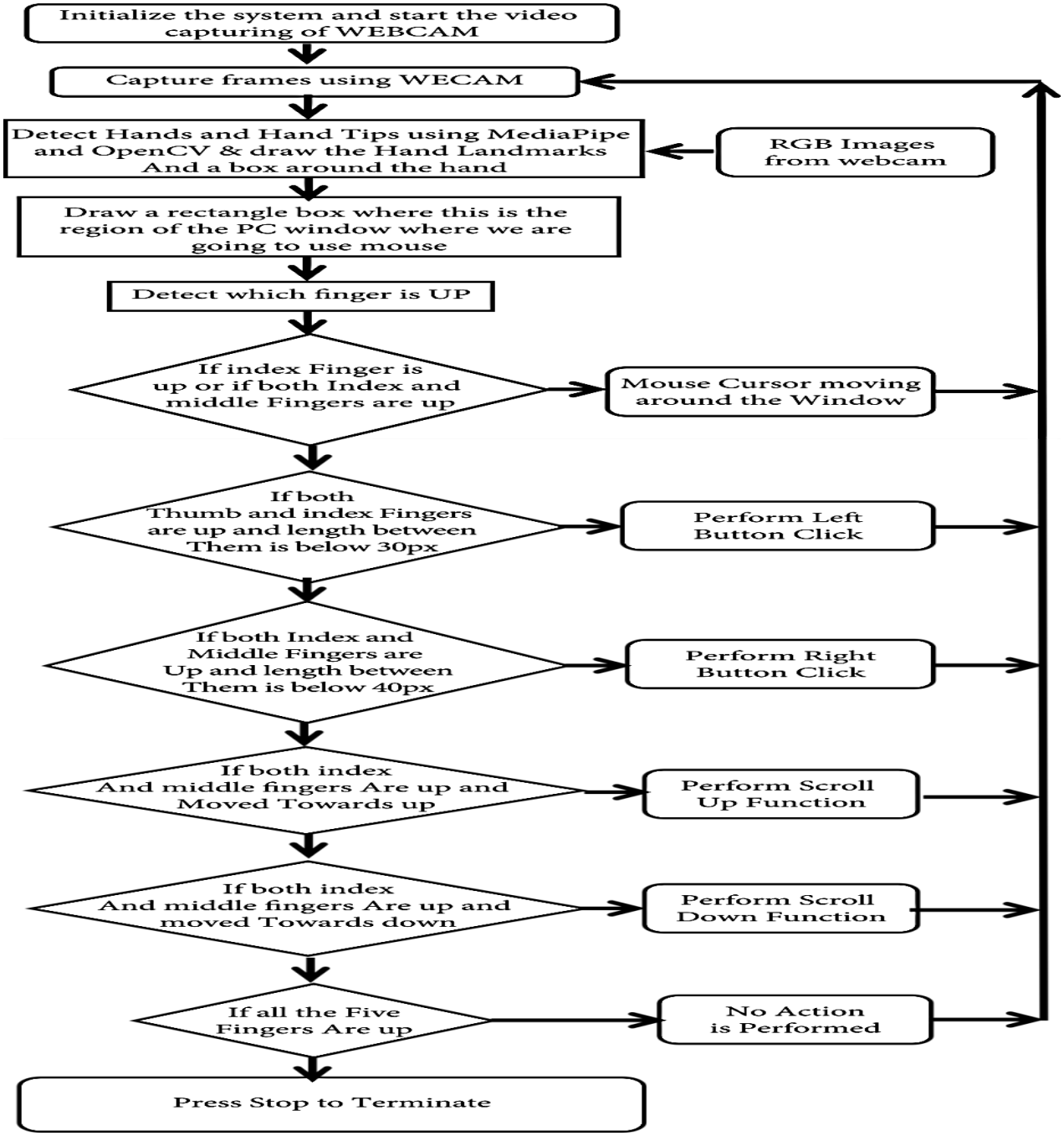


Figure:- 7.1

Data flow diagramming is a graphical technique that depicts the information flow and the transforms that are applied as data move from input to output. They are part of the structure model in the software development life cycle.

7.2 Basic Data Flow Notation :-

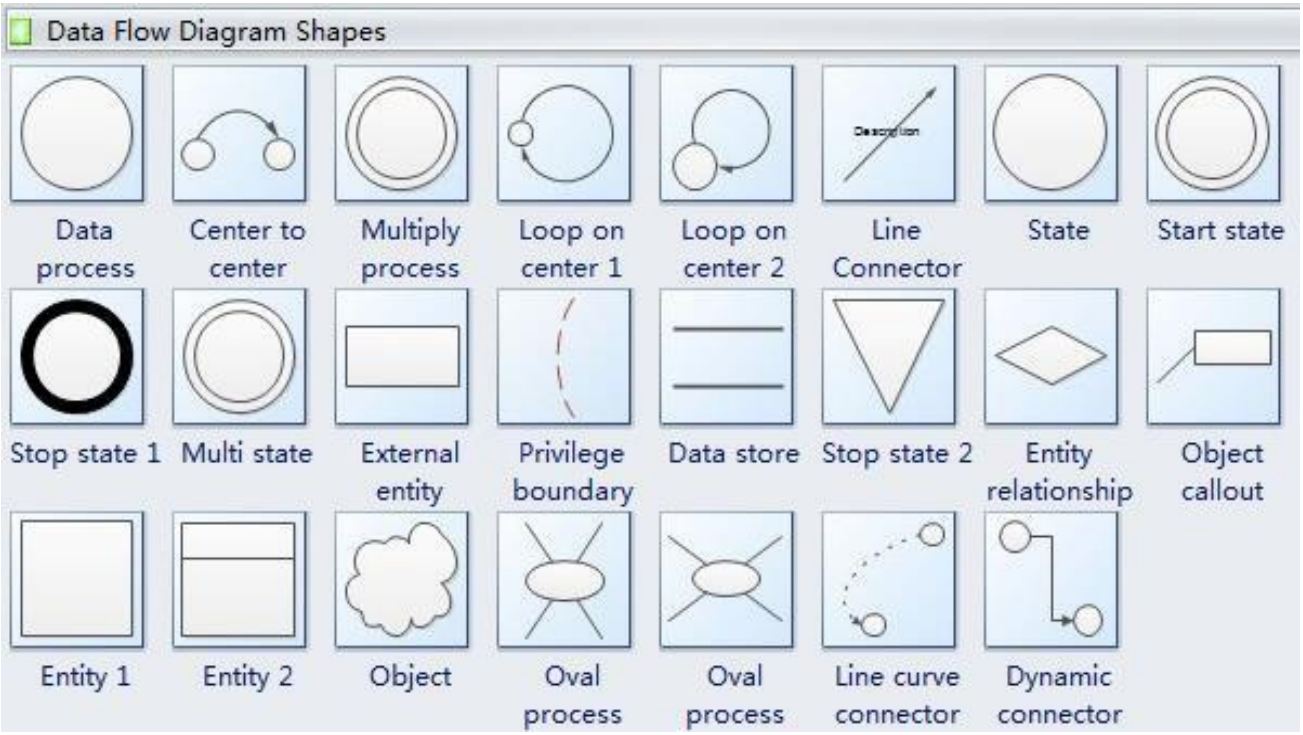


Figure:- 7.2

7.3 Constructing a DFD :-

Rules for drawing DFD's :

- Process should be named and numbered for easy reference. Each name should be representative of the process.
- The direction of flow is from top to bottom and from left to right. Data traditionally flows from source to destination although they may flow back to the source. One way to indicate this is to draw a long flow line back to a source. An alternative way is to repeat the source symbol as a destination. Since it is used more than once in the DFD it is marked with a short diagonal.
- When a process explodes into lower level details, they are numbered.
- The names of data stores and destinations are written in capital letters. Process and data flow names have the first letter of each work capitalized.

7.4 Silent Features of DFD's :-

The Various key features of DFD's are as follows :

- The DFD shows flow of the data not to control loops and decision re controlled considerations do not appear on a DFD
- The DFD does not indicate the time factor involved in any process whether the data flows take place daily, weekly, monthly or yearly.
- The sequence of events is not brought out on the DFD
- Information and/or data flow is represented by a labeled arrow.
- The data-flow diagram is part of the structured-analysis modeling tools
- DFD is a system created by analysts based on interviews with system users

7.5 DATA DICTIONARY :-

The data dictionary is used to create and store definitions of data, location, format for storage and other characteristics. The data dictionary can be used to retrieve the definition of data that has already been used in an application. The data dictionary also stores some of the description of data structures, such as entities, attributes and relationships. It can also have software to update itself and to produce reports on its contents and to answer some of the queries.

They are:

- Asking or interviewing
- Determining from the existing system
- Analyzing the critical success factors
- Experimentation and modeling.

7.6 FUNCTIONAL AND PERFORMANCE SPECIFICATIONS :-

Functional Performance Specification (FPS) is a technique to define the requirements of a project, product or service, based on the required Functions and the specific needs related to those Functions. The Functional Performance Specification (FPS) technique enables the owner to document their performance requirements in terms of functions and needs without reference to a specific solution. For each function, needs are expressed in terms of assessment criteria, levels of performance and a flexibility for each level

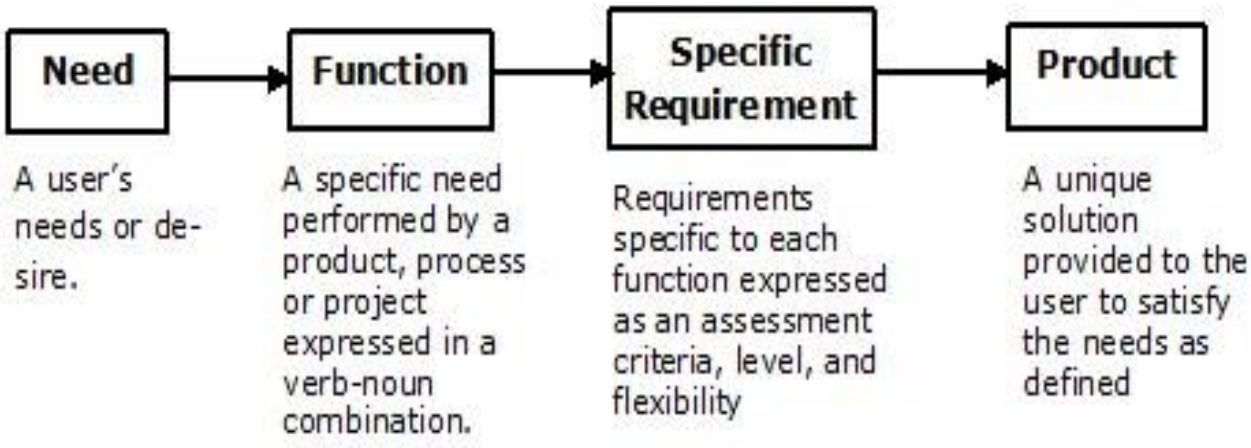


Figure 7.6

Using the FPS technique, Needs and expected performance are expressed without reference to a specific solution. The client defines the needs in terms of what the solution must **do**, rather than what it must **be**. The Functional Performance Specification document provides a common understanding of project needs between the client and the supplier and it allows for a broader range of innovative solutions that meet the client needs. Documenting needs through the FPS technique allows the owner to obtain innovative solutions that best meet the actual requirements.

The development of an FPS encourages constructive dialogue with the client and enables the provider to express greater creativity in the delivery of solutions. Using this technique to document needs changes the owner/supplier relationship from an obligation to respond to a predetermined technical specification to an obligation to achieve a result.

7.6.1 Use of FPS Technique :-

- Defining project goals and requirements.
- Identifying business needs to develop or improve business processes ● Restructuring organizations.
- Defining requirements for Design Build projects.

CHAPTER-8 HAND GESTURES FUNCTIONALITY

8.1 SCREENSHOTS OF HAND DETECTION :-

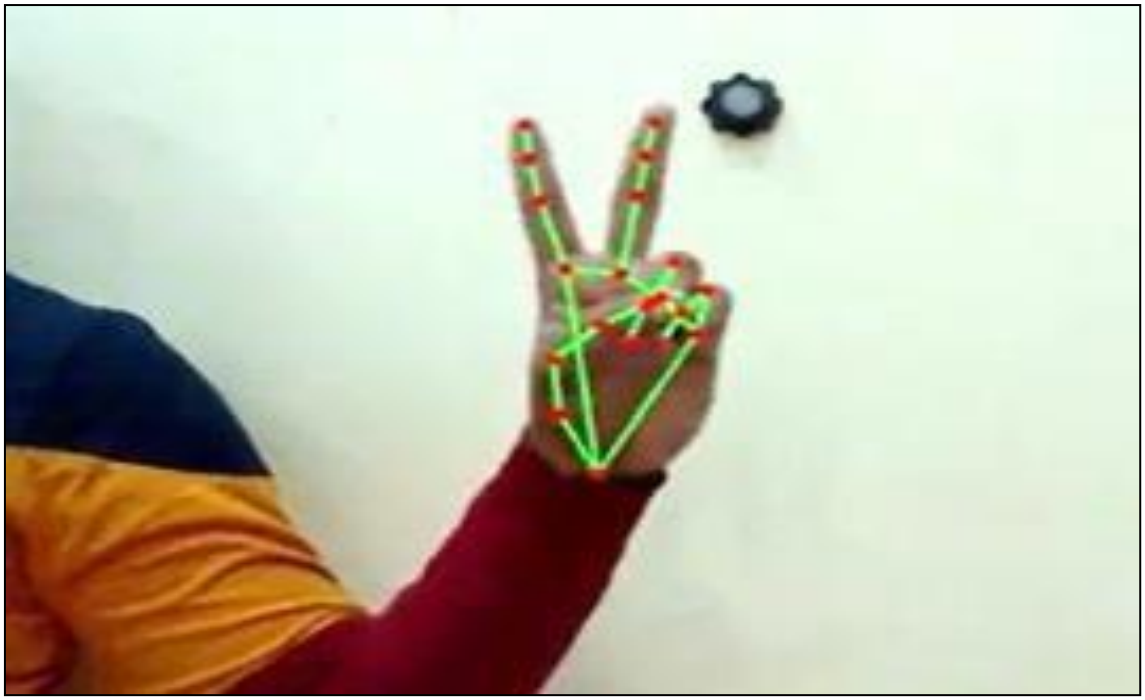


Figure:-8.1

8.2 SCREENSHOTS OF PALM GESTURE :-

Figure:-8.2

8.3 SCREENSHOTS OF CURSOR MOMENT GESTURE :-

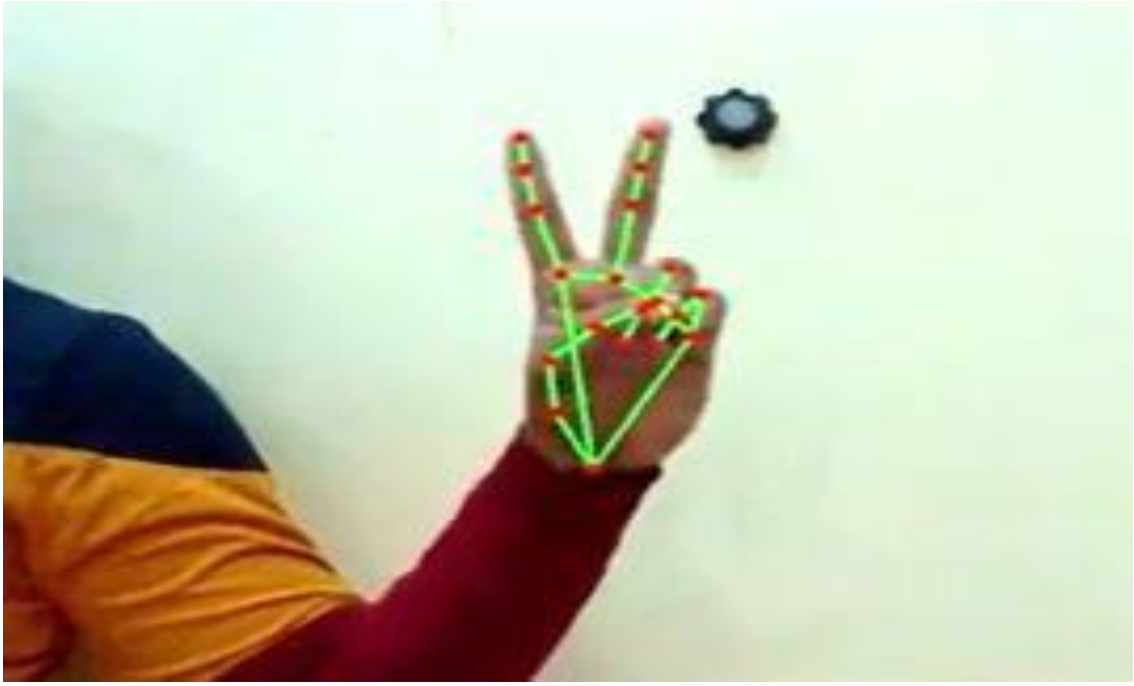


Figure:-8.3

8.4 SCREENSHOTS OF RIGHT CLICK GESTURE :-

Figure:-8.4

8.5 SCREENSHOTS OF LEFT CLICK GESTURE :-

Figure:-8.5

8.6 SCREENSHOTS OF DOUBLE CLICK GESTURE :-

Figure:-8.6

8.7 SCREENSHOTS OF DRAG AND DROP GESTURE :-

Figure:-8.7

8.8 SCREENSHOTS OF BRIGHTNESS CONTROL GESTURE :-

Figure:-8.8

8.9 SCREENSHOTS OF VOLUME CONTROL GESTURE :-

Figure:-8.9

8.10 SCREENSHOTS OF SCROLLING GESTURE :-

Figure:-8.10

CHAPTER-9 TESTING

Once the coding phase is completed, the testing phase takes place. The testing process focuses on the conducting tests to uncover errors and to ensure that the given input will produce the desired output. The project has been tested on raw data.

9.1 TEST STRATEGY :-

- **Unit Testing** : - In this individual modules are tested differently. We apply the different test cases for right output that are following define:
    - **Test case 1**: We test the login form with different inputs given to the form. In the first case the right username and password are entered by the user/admin in form and submit. The message will be displayed if thelogin is successful.
    - **Test case 2**:In the second case user/admin enter the wrong user name and password then submit. Now the error message print that the username and password doesn't match.

- **Integration Testing** : - Integration testing is a systematic technique for testing the different modules together. The different types of inputs given to the integration system and checked out whether the right output will come or not.

- **System Testing** : - System testing in which the whole system developstests to uncover the errors. Each and every element of the websiteis tested whether all pages work properly or not.

- **Acceptance Testing**: - Acceptance testing is performed at the client side to ensure the user has accepted the system.

- **Functional Testing :-** In this testing the validations are tested whether they work correctly or not. The login and sign up forms are tested for validations.

- **Usability Testing:-**The usability testing applies on the content of the website which the developer makes usable to the user. The content like images, videos etc that are viewed by the user.

- **Compatibility Testing :-** The compatibility testing is the testing about the compatibility of the developed system regarding the supportive web browsers and operating systems.

- **Database Testing:-** In database testing we test the queries that are used to insert, delete, update the data in the database. Whether the data is properlystored in or fetched from the database or not.

- **Link Testing:-** The link testing is used to test the links that are used for navigation from one page to another page

9.2 TEST CASES DESIGNED :-

The primary objective of test case design is to derive a set of tests that are used for uncovering errors in software. The test case specification is the major activity in the testing process. Careful selection of test cases that satisfy the requirements is essential for proper testing. Various characteristics of test cases that are required for portal are:

- A good test has a high probability of finding an error.
- A good test is not redundant.
- A good test should be neither too simple nor too complex.

9.2.1 Test Results :-

- Actual Results: pass / failed.
- If passed –What actually happens when you run the test.
- If failed – put in description of what you've observed.

  Once code has been generated, the project must be tested to uncover as many errors as possible before delivery to the customer. The testing process ensures that all statements

have been tested, and on the functional elements that are conductingtests to uncover errors and ensures that defined input will produce actual results that agree with required results.

The table shows that more than half the errors one can have in a system are introduced during requirements analysis. On the other hand, one only discovers a minority (5%) of the errors in a system during requirement analysis. Another point to be noted is that 55% of the faults are introduced in the first phase of the software development project.

CHAPTER-10 IMPLEMENTATION

10.1 TOOLS USED FOR IMPLEMENTATION :- WINDOW'S WINDOWS-10 LANGUAGE PYTHON (WITH OPENCV AND MEDIAPIPE LIBRARIES)

Implementation involves placing the completed and tested system of hardware and software into the actual work environment of the users. As the system is tested it starts to move into the implementation phase. Ideally the system should be completed and fully tested before implementation gets under way.

As soon as the first phase of implementation- file set-up-starts, all system documentation should be available, user manuals, procedures manuals, computer operating instructions and security procedures. The system then passes from the development staff to the computer operations personnel and once the system is live, strict procedures should be enforced governing programmer's access to programs and files.

There are both technical & people oriented activities during this stage.

Figure:- 10.1

10.2 FUTURE SCOPE AND CONCLUSION :-

Due to accuracy and efficiency plays an important role in making the program as useful as an actual physical mouse, a few techniques had to be implemented. After implanting such type of application there is big replacement of physical mouse i.e., there is no need of any physical mouse. Each &

every movement of physical mouse is done with this motion tracking mouse (virtual mouse). There are several features and improvements needed in order for the program to be more user friendly, accurate, and flexible in various environments. The following describes the improvements and the features required:

a) **Smart Movement:-** Due to the current recognition process are limited within 25cm radius, an adaptive zoom in/out functions are required to improve the covered distance, where it can automatically adjust the focus rate based on the distance between the users and the webcam.

b) **Better Accuracy & Performance:-** The response time are heavily relying on the hardware of the machine, this includes the processing speed of the processor, the size of the available RAM, and the available features of webcam. Therefore, the program may have better performance when it's running on a decent machine with a webcam that performs better in different types of lightings.

c) **Mobile Application:-** In future this web application also able to use on Android devices, where touchscreen concept is replaced by hand gestures.

REFERENCES

1. OpenCV Website →www.opencv.org
2. MSDN Microsoft developers network – www.msdn.microsoft.com
3. https://www.tutorialspoint.com› → opencv.
4. https://mediapipe.dev/ → mediapipe
5. https://pypi.org/project/PyAutoGUI/ → PyAutoGUI
6. https://pypi.org/project/autopy/ → AutoPy
7. https://pypi.org/project/pynput/ → Pynput
8. https://www.hindawi.com/journals/jhe/2021/8133076/ → Documentation